Fiscal Year:	FY 2010	**Task Last Updated:**	FY 01/26/2010
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	Comparative Analysis of Charged Particle-Induced Autosomal Mutations in Murine Tissues and Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	a_kronenberg@lbl.gov	**Fax:**	FY 510-486-4475
PI Organization Type:	GOVERNMENT	**Phone:**	510-486-6449
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	1 Cyclotron Road, Building 977-180		
PI Address 2:			
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	04/01/2007	**End Date:**	03/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Project extended until 3/31/2012 per C. Guidry/JSC (Ed., 2/14/2011)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Turker, Mitchell (Oregon Health & Science University)		
Grant/Contract No.:	NNJ07HC72I		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Mutations are required in the transition from normal cells to cancer cells. Solid cancers in people arise primarily in epithelial cells. This proposal addresses the carcinogenic risk of exposure to energetic protons, the most abundant ion in galactic cosmic radiation (GCR) and solar particle events. The toxic effects of these exposures will also be measured. The studies outlined quantify mutation induction at a model, autosomal genetic locus (Aprt) located on mouse chromosome 8. C57BL/6 x DBA2 F1 Aprt heterozygous mice or kidney epithelial cells from this mouse strain will be irradiated to determine dose responses for cell killing and Aprt mutation in kidney epithelial cells exposed in the mice, or in kidney epithelial cells irradiated in culture. This will allow assessment of the role of the tissue microenvironment on modulation of mutation risk and on persistence of radiation-induced mutants in the kidney. Molecular analysis will determine the mechanisms leading to proton-induced Aprt mutants in exposed kidneys or in kidney cells irradiated in		

Task Description:	culture. It is predicted that proton exposure will result in mutants that lose the Aprt gene and other genes that are located in nearby regions of mouse chromosome 8. Studies are proposed to test if protracted exposure lessens the frequency of proton-induced mutants while also lowering the risk of killing kidney epithelial cells. The toxic and mutagenic effects of sequential exposure to protons and iron ions will be assessed (with iron ions serving as a representative heavy ion in the GCR) to test whether the risks are additive. Results from these studies will help reduce uncertainties in estimating the risk of cancer for astronauts exposed to protons alone or in mixed radiation fields. They will also provide detailed knowledge of the types of mutations that are created by these exposures that can contribute to the development of cancer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will have a direct impact on our understanding of proton-induced toxicity and mutations in a mammalian epithelial tissue. This has relevance to cancer therapy involving proton exposures that not only target malignant tissue, but which also involve limited exposure of normal tissues that can develop late complications as a result of proton exposure. In addition, our in vivo/in vitro approach also illuminates the importance of the normal tissue environment on the modulation of radiation sequelae.
Task Progress:	During the Spring 2009 run at BNL, replicate dose-response curves were performed for mouse kidney epithelial cell mutagenesis at the endogenous, autosomal heterozygous Aprt locus in cells exposed to 1 GeV protons in vitro. The results indicate a dose-dependent increase in mutation for acute exposures from 0-5 Gy, confirming the results of the initial experiments performed in 2008. These studies assess the early mutagenic response to an important class of charged particle radiation found in spaceflight environments in an epithelial model derived from B6D2F1 mice. Parallel studies continued for B6D2F1 mouse kidney epithelial cells for acute exposures to graded doses of 1 GeV protons in situ. Preliminary analyses demonstrate consistency in the background mutant fractions across different run years and harvest times. Data collection is ongoing for the final cohort and a definitive statistical analysis of both cytotoxicity data and mutant frequency data awaits the completion of these studies. Analysis of proton-induced Aprt mutational spectra continues for B6D2F1 mouse kidney epithelial cells. Eighty independent Aprt-deficient mutants were isolated after in vitro exposure of kidney epithelial cells to 5 Gy of 1 GeV protons. A PCR-based analysis of mutational spectra was performed for changes along mouse chromosome 8. An initial comparison with a series of independent, spontaneously arising mutants indicates a difference between the spontaneous mutants and those arising after proton exposure ($p<0.001$). A large series of proton-induced mutants have been collected from primary cultures of B6D2F1 kidney epithelial cells established after in situ exposures harvested at early and late harvest times. The results of these studies are ongoing. The results from this grouping will be compared against the results of obtained for the 80 B6D2F1 kidney epithelial cells exposed in vitro to evaluate the impact of the microenvironment on proton-induced genetic alterations.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Kronenberg A, Sudo H, Dan C, Sandfort K, Turker M. "Evidence for genomic instability in normal mouse epithelium in situ and human mammary epithelial cells exposed to Fe ions in three-dimensional acinar structures." Presented at the Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009. Proceedings of the Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009, page 217. , Jul-2009
Abstracts for Journals and Proceedings	Zafar F, Seidler SB, Kronenberg A, Schild D, Wiese C. "Homologous recombination contributes to the repair of DNA double-strand breaks induced by 1 GeV/n Fe ions." Presented at the Heavy Ions in Space Symposium 2009, Cologne, Germany, July 6-10, 2009. Proceedings of the Heavy Ions in Space Symposium 2009, page 111. , Jul-2009
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Articles in Peer-reviewed Journals	Kronenberg A, Gauny S, Kwoh E, Connolly L, Dan C, Lasarev M, Turker MS. "Comparative analysis of cell killing and autosomal mutation in mouse kidney epithelium exposed to 1 GeV/nucleon iron ions in vitro or in situ." Radiation Research 2009 Nov;172(5):550-7. PubMed PMID: 19883222 , Nov-2009
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